

```
ghci> rs <- makeResolvSeed defaultResolvConf
```

```
ghci> withResolver rs $ \resolver -> lookupA resolver hostname
Right [100.162.203.60,100.162.204.60]
```

The *defaultResolvConf* is of type *ResolvConf* and consists of the following default values:

```
-- * 'resolvInfo' is 'RCFilePath' \"\\\"/etc\\resolv.conf\".
--
-- * 'resolvTimeout' is 3,000,000 micro seconds.
--
-- * 'resolvRetry' is 3.
```

The *makeResolvSeed* and *withResolver* functions assist in making the actual DNS resolution. The *lookupA* function obtains all the A records for the DNS entry. Their type signatures are shown below:

```
ghci> :t makeResolvSeed
makeResolvSeed :: ResolvConf -> IO ResolvSeed
```

```
ghci> :t withResolver
withResolver :: ResolvSeed -> (Resolver -> IO a) -> IO a
```

```
ghci> :t lookupA
lookupA
:: Resolver
-> dns-1.4.5:Network.DNS.Internal.Domain
-> IO
(Either
 dns-1.4.5:Network.DNS.Internal.DNSError
 [iproute-1.4.0:Data.IP.Addr.IPV4])
```

The *lookupAAAA* function returns all the IPv6 'AAAA' records for the domain. For example:

```
ghci> withResolver rs $ \resolver -> lookupAAAA resolver
hostname
Right [2400:cb00:2048:1::6ca2:cc3c,2400:cb00:2048:1::6ca2:cb3c]
```

Its type signature is shown below:

```
lookupAAAA
:: Resolver
-> dns-1.4.5:Network.DNS.Internal.Domain
-> IO
(Either
 dns-1.4.5:Network.DNS.Internal.DNSError
 [iproute-1.4.0:Data.IP.Addr.IPV6])
```

The MX records for the hostname can be returned using the *lookupMX* function. An example for the *shakthimaan.com*

website is as follows:

```
ghci> import Network.DNS.Lookup
ghci> import Network.DNS.Resolver
```

```
ghci> let hostname = Data.ByteString.Char8.pack "www.shakthimaan.com"
```

```
ghci> rs <- makeResolvSeed defaultResolvConf
```

```
ghci> withResolver rs $ \resolver -> lookupMX resolver
hostname
Right [(("shakthimaan.com.",0)]
```

The type signature of the *lookupMX* function is as follows:

```
ghci> :t lookupMX
lookupMX
:: Resolver
-> dns-1.4.5:Network.DNS.Internal.Domain
-> IO
(Either
 dns-1.4.5:Network.DNS.Internal.DNSError
```



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